Evaluating the Impact of Narrative-Based Augmented Reality on Early STEAM Learning Outcomes in Sri Lankan Primary Education

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Primary school students in Sri Lanka consistently struggle to understand biological processes in the Grade 4–5 Parisaraya curriculum due to heavy reliance on textbook-based instruction, which cannot represent invisible phenomena such as seed germination, food chain energy transfer, or ecosystem interdependence. This proposal presents Component 2 of a broader four-component group research project: an Augmented Reality (AR) Life Science Storybook designed to address this gap. The proposed system uses a Unity and Vuforia mobile AR application paired with a printed bilingual storybook to bring biological concepts to life through AR visualizations triggered by narrative storybook pages. Students interact with plant life cycles, food chains, and ecosystem processes through culturally grounded Sri Lankan species and settings. The study employs a quasi-experimental design to evaluate learning outcomes and knowledge retention in comparison with conventional textbook instruction. Expected outcomes include improved conceptual understanding, increased engagement, and a validated AR framework for primary science education in Sri Lanka.

Keywords: Augmented Reality, Life Science Education, AR Storybook, Primary Education, STEAM Learning, Arts Integration, Sri Lanka

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LIST OF ABBREVIATIONS

Abbreviation	**Definition**
AR	Augmented Reality
STEM	Science, Technology, Engineering, and Mathematics
STEAM	Science, Technology, Engineering, Art, and Mathematics
TAF	Topic Assessment Form
SDG	Sustainable Development Goal
IM	Interactive Media
SDK	Software Development Kit
SLIIT	Sri Lanka Institute of Information Technology

1. INTRODUCTION

1.1 Background and Context

The Sri Lankan primary education system faces a well-documented challenge in delivering effective STEM instruction at the Grade 4 and Grade 5 levels. National examination data from the Grade 5 Scholarship Examination consistently reveals that students perform poorly in topics requiring visual, spatial, and process-based reasoning, particularly in mathematics and science [1]. The subjects of measurement, life science, physical science, and number concepts are repeatedly identified as the weakest performance areas, reflecting a systemic gap between how concepts are taught and how children at ages 9 to 11 actually learn [2].

Current classroom instruction in Sri Lanka predominantly relies on printed textbooks and verbal teacher explanation. While these methods convey factual content, they are fundamentally limited in their ability to make abstract and dynamic concepts concrete for young learners. A textbook diagram of seed germination, for instance, shows a static cross-section but cannot animate the sequence of root growth, shoot emergence, and leaf development. A printed food chain diagram labels organisms but cannot show the dynamic energy flow between a producer, a primary consumer, and a decomposer. Children are required to memorize labels and stages without ever experiencing the underlying processes in a meaningful way [3].

Life Science, taught under the Grade 4 and 5 Parisaraya curriculum, is among the subjects most severely affected by these pedagogical limitations. The narrative storybook format naturally integrates artistic elements such as illustrated characters, culturally grounded visual storytelling, and bilingual creative writing, positioning Art as a meaningful bridge between abstract biological concepts and student engagement within the STEAM education framework. Biological phenomena such as germination, decomposition, food chains, and ecosystem water cycles are entirely invisible in a standard classroom environment. Students encounter these concepts through simplified diagrams and descriptive sentences, which consistently shows produce surface-level memorization rather than durable conceptual understanding [4].

Globally, Augmented Reality (AR) has emerged as a powerful educational technology that bridges the gap between abstract representation and experiential understanding. AR overlays interactive digital content onto the physical world in real time, allowing students to observe dynamic processes, manipulate virtual objects, and engage with subject matter in a spatially and contextually grounded manner [5]. Empirical studies across multiple national contexts have demonstrated that AR-based learning environments improve conceptual understanding, increase student engagement, and support knowledge retention significantly better than static instructional media [6].

Despite this growing global evidence base, no AR-based learning system exists for the Sri Lankan primary science or mathematics curriculum. The existing digital education resources available to Sri Lankan schools are minimal, concentrated in urban areas, and do not incorporate interactive or immersive technologies. No bilingual Sinhala and English AR educational content has been developed for any primary school subject in Sri Lanka, and the use of narrative-driven AR for children in this age group has not been studied in any South Asian primary school context [7].

The broader research project of which this component forms a part addresses this gap through a four-component approach. The overall project, titled Evaluating the Impact of Narrative-Based Augmented Reality on Early STEM Learning Outcomes in Sri Lankan Primary Education, develops four independent AR storybook systems covering measurement and data handling, life science, physical science, and number concepts. Each component is designed as a standalone research study with its own storybook, AR application, and quasi-experimental evaluation. This proposal documents Component 2: the AR Life Science Storybook.

This component aligns directly with the United Nations Sustainable Development Goal 4 (Quality Education), which calls for inclusive and equitable quality education and the promotion of lifelong learning opportunities for all. By providing an immersive, bilingual, culturally grounded learning experience for underserved primary students in Sri Lanka, this project contributes directly to improving educational quality and equity at the grassroots level.

1.2 Literature Review

The application of Augmented Reality in primary and secondary science education has been studied extensively over the past decade. The literature consistently supports the view that AR environments improve learning outcomes, engagement, and knowledge retention compared to conventional instruction, particularly in subjects that require visualization of dynamic or invisible phenomena.

Antoniadi (2023) conducted a case study with Grade 1 primary students using an AR application designed to teach plant anatomy. The study reported significant improvements in students' ability to identify and name plant parts following AR instruction compared to textbook-based teaching. The interactive nature of the AR environment was identified as a key factor in sustaining student attention and supporting recall [8].

Permana et al. (2024) investigated primary students' engagement with AR-based content covering plant development cycles. Their findings demonstrated that AR significantly increased both engagement and conceptual retention compared to static instructional materials, with students able to describe the sequence of plant growth stages more accurately following AR interaction [9].

Meilinda et al. (2024) conducted a systematic review of 42 studies on collaborative learning in biology through AR, published in SULE-IC 2024. The review concluded that AR consistently outperforms traditional instruction in supporting understanding of biological processes that are not directly observable in a classroom, including cellular structures, organismal development, and ecological relationships [10].

Zammit et al. (2025) published a literature review of 40 empirical studies on AR in biology education, with a focus on Unity and Vuforia-based implementations. Their analysis confirmed that AR-based biology learning environments produce stronger pre- to post-test gains than control groups using conventional methods, and that the effect is particularly pronounced for process-based topics such as life cycles, decomposition, and energy flow in food chains [11].

Arici et al. (2024) conducted a systematic literature review and meta-analysis of AR in natural sciences and biology teaching, analyzing studies published between 2014 and 2024. The meta-analysis reported a statistically significant overall effect size favouring AR over traditional instruction, with process visualization identified as the primary mechanism of improvement [12].

Kuhn et al. (2025) presented a scoping review of AR research in environmental education, identifying a strong evidence base for AR-supported understanding of ecosystem relationships and environmental interdependence. The review highlighted the persistent lack of AR tools designed for non-Western, bilingual, or curriculum-specific contexts [13].

Nevrelova et al. (2024) examined the use of AR for enhancing digital literacy and science understanding in primary education, reporting that students exposed to AR-based lessons showed significantly greater ability to apply biological knowledge to real-world scenarios compared to textbook-only groups [14].

Wu et al. (2024) conducted a systematic literature review of AR in early childhood and elementary education, covering studies from 2015 to 2024. Their findings confirmed AR as an effective tool for improving learning outcomes in STEM subjects, with life science and environmental science identified as the domains with the most consistently positive effects [15].

Collectively, this body of literature establishes a strong empirical foundation for the proposed AR Life Science Storybook. However, a critical gap exists: no study in this literature has developed or evaluated an AR life science learning system for Sri Lankan primary students, used local biodiversity and culturally grounded settings as the narrative basis for AR content, or deployed a narrative-driven storybook as the primary interface for triggering AR in a South Asian primary school context.

1.3 Research Gap

The primary research problem of this group project is that Sri Lankan Grade 4 and 5 students consistently underperform in STEM subjects, particularly in topics that require visual, spatial, and process-based reasoning. The existing instructional infrastructure limited to printed textbooks and verbal explanation is fundamentally unable to represent the dynamic, invisible, and three-dimensional nature of the concepts these students are expected to understand. This pedagogical gap is well-documented in national examination performance data and has persisted without a technology-based intervention designed specifically for the Sri Lankan primary curriculum [1], [2].

Within Life Science education, this gap is particularly acute. The Grade 4 and 5 Parisaraya curriculum requires students to understand processes seed germination happening underground over days, energy transferring invisibly between organisms in a food chain, decomposers breaking down organic matter at a microscopic level that are by definition impossible to observe directly in a classroom. Current instruction provides textbook diagrams and descriptive labels, which produce memorization of terminology without any meaningful understanding of the biological process the terminology describes [4], [7].

Three specific gaps define the research problem for this component. First, no AR-based life science learning tool has been developed for any primary school subject in the Sri Lankan curriculum. Second, no AR life science educational content exists in Sinhala, the mother tongue of the majority of Sri Lankan primary students, making all existing AR biology tools linguistically inaccessible to the target population. Third, no study has investigated whether a narrative-driven AR approach in which the storybook narrative creates a meaningful context before the AR visualizes the biological process produces superior learning outcomes compared to conventional or marker-only AR instruction in a South Asian primary school setting [13], [15].

These three gaps curricular, linguistic, and pedagogical collectively justify the development and evaluation of the proposed AR Life Science Storybook.

1.4 Main Research Question:

Can a narrative-based mobile AR system improve STEAM concept understanding and knowledge retention in Sri Lankan Grade 4-5 students across mathematics and science topics compared to conventional textbook-based instruction?

1.5 Sub-Research Questions:

SRQ1: Does a narrative-driven AR measurement experience improve conceptual understanding and practical application accuracy for measurement topics (length, mass, capacity, time, and data) among Sri Lankan Grade 4-5 students compared to conventional textbook-based instruction?

SRQ2: Does a narrative AR storybook depicting biological life cycles and ecosystem processes improve scientific process understanding and knowledge retention in Sri Lankan Grade 4-5 students compared to static diagram-based textbook instruction?

SRQ3: Does an interactive AR physical science storybook improve conceptual understanding of light, shadow, and forces and reduce persistent misconceptions in Sri Lankan Grade 4-5 students compared to teacher-led demonstration with static diagrams? (This proposal focuses on SRQ3.)

SRQ4: Does a narrative AR storybook embedding fraction, decimal, and number operation concepts in a real-world context improve fraction conceptual understanding and number operation accuracy in Sri Lankan Grade 4-5 students compared to conventional symbolic instruction?

2. OBJECTIVES

2.1 Main Objective:

To design, develop, and evaluate a set of four narrative-based mobile Augmented Reality storybook applications that improve STEAM concept understanding and knowledge retention among Sri Lankan Grade 4-5 students across mathematics and science topics, compared to conventional textbook-based classroom instruction.

2.2 Specific Objectives:

The four components of the broader research project address the following sub-questions:

- To develop a narrative-driven AR measurement storybook application that makes length, mass, capacity, time, and data concepts visually concrete for Grade 4-5 mathematics students through interactive 3D measurement instrument simulations. (Component 1)

- To develop a narrative-driven AR life science storybook application that shows biological processes plant life cycles, food chains, and ecosystems using locally grounded Sri Lankan species and integrates artistic visual storytelling through illustrated bilingual narrative pages, to improve scientific understanding and creative engagement in Grade 4-5 Parisaraya students. (Component 2)

- To develop a narrative-driven AR physical science storybook application that makes light ray propagation, shadow formation, and force dynamics spatially visible and interactively manipulable for Grade 4-5 Parisaraya students, targeting the misconceptions that arise from static, diagram-only instruction. (Component 3 this proposal)

- To develop a narrative-driven AR number concepts storybook application that embeds fractions, decimals, place value, and number operations within a real-world bakery context to improve number concept understanding and operational accuracy in Grade 4–5 mathematics students. (Component 4)

Component 2 - the AR Life Science Storybook - is the focus of this proposal.

3. METHODOLOGY

3.1 Research Approach

This study adopts a quasi-experimental research design with pre-test and post-test measurements. The design is appropriate because random assignment of participants is not feasible within an operational primary school environment, yet controlled comparison between an experimental group and a control group is essential to evaluate the intervention's effect on learning outcomes. The experimental group will use the AR Life Science Storybook, while the control group will receive the equivalent content through conventional textbook-based instruction. Both groups will be drawn from the same school, with matched cohorts selected to minimize demographic and academic differences.

3.2 Technical Methods and Development Approach

The system will be developed using Unity 2022 LTS as the game development and AR application framework, with Vuforia Engine as the AR software development kit for image target detection and tracking. Each page of the printed storybook serves as a Vuforia ImageTarget marker. On scanning, Unity renders a three-dimensional AR scene anchored to the page surface, which follows the narrative stage of that page.

The development will follow a Design Science Research (DSR) methodology [1], iterating through three phases: (1) problem identification and requirements definition; (2) AR system design, asset creation, and application development; and (3) evaluation through quasi-experimental study. Three-dimensional biological models will be created using Blender and imported into Unity. AR visualizations and interactive AR scenes will be implemented using Unity and controlled by a Narrative Manager script. that sequences AR events based on the storybook page being detected. All textual content will be bilingual in Sinhala and English.

3.3 Tools and Platforms

Unity 2022 LTS is selected as the development platform because it offers robust Vuforia integration, Android mobile AR deployment, and a mature 3D animation pipeline. Vuforia is selected for its proven performance as an image target tracking SDK in mobile AR educational applications. Blender is used for 3D asset creation because it is open-source, offers professional-grade modelling and animation capabilities, and produces Unity-compatible export formats. The application will target Android devices running version 8.0 (Oreo) and above, which represent the most affordable and widely available devices in Sri Lankan schools.

3.4 Data Collection and Ethical Considerations

Data will be collected from Grade 4 and 5 students at a selected primary school in Sri Lanka. Pre-test assessments will be administered to both groups to establish baseline knowledge of the relevant life science topics. Post-test assessments using the same instrument will be administered following the intervention period. Assessment instruments will be designed in Sinhala and English, reviewed by subject matter experts, and validated through pilot testing before full deployment.

Ethical approval will be obtained from the SLIIT Faculty Research Ethics Committee prior to any data collection. Written informed consent will be obtained from the parents or guardians of all participant children. School administration approval will be secured, and all student data will be anonymized and stored securely in accordance with applicable data protection regulations. Participation will be entirely voluntary, with no academic consequence for withdrawal.

3.5 Validation Strategy and Performance Metrics

The primary performance metrics are: (1) pre-test to post-test knowledge gain scores measured by a validated instrument; (2) knowledge retention scores measured by a delayed post-test administered two weeks after the intervention; and (3) student engagement measured through structured observation and a validated engagement scale adapted for primary school students. Statistical analysis will use independent samples t-tests to compare mean scores between groups, with effect size calculated using Cohen's d. A significant threshold of $p < 0.05$ will be applied throughout.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overall AR Learning System Architecture

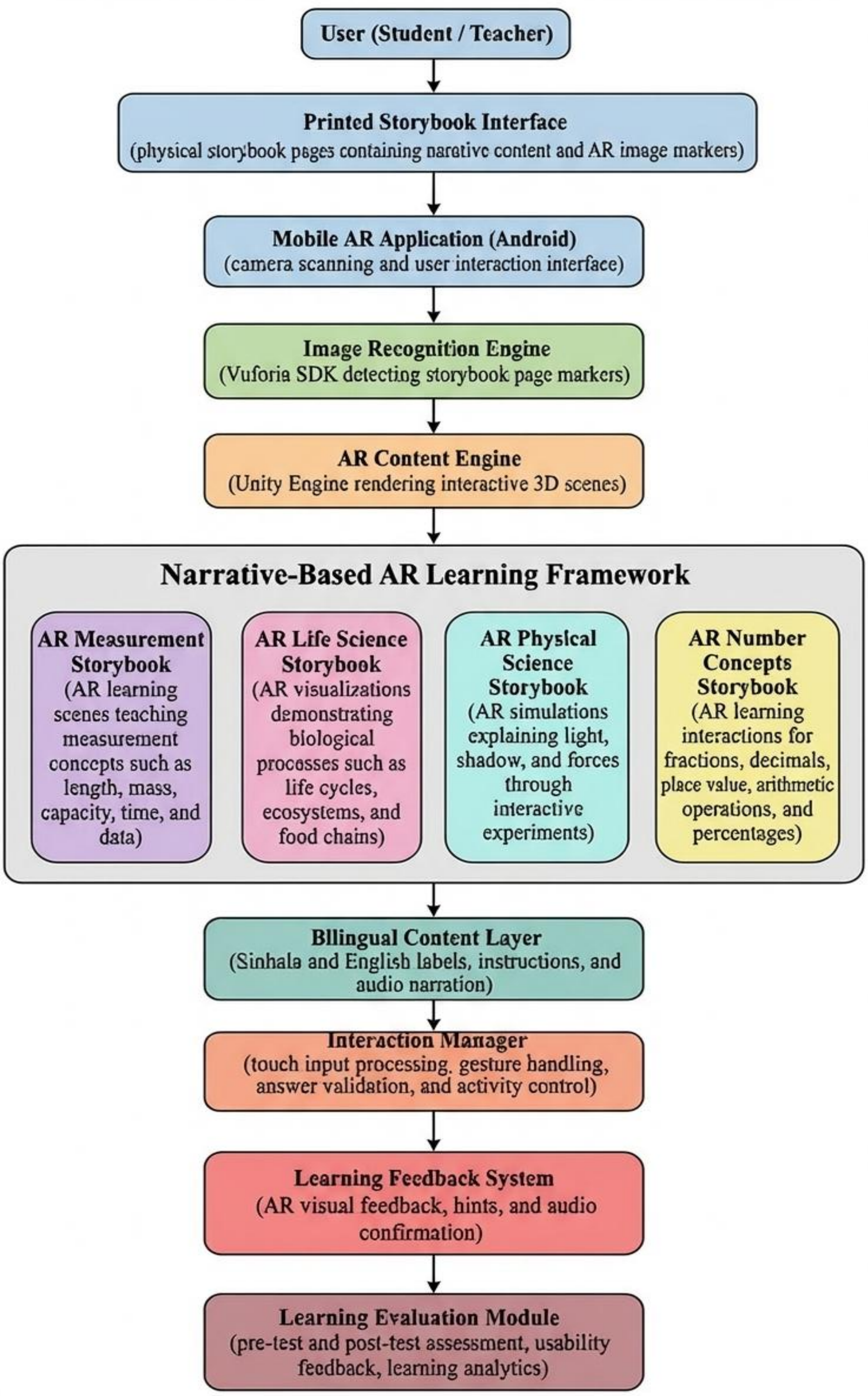

Figure 1: Overall AR Learning System Architecture

The broader AR learning system for this research project consists of four independent component applications that share a common architectural pattern. Each component operates as a standalone mobile AR application built on Unity and Vuforia, requiring no network connectivity during student use, which is critical for deployment in schools with limited infrastructure.

At the top level, the system comprises four layers: (1) the Physical Interface Layer, consisting of the printed bilingual storybook with embedded Vuforia Image Target markers on each page illustration; (2) the Mobile Application Layer, the Unity and Vuforia application running on the student's Android device; (3) the Content Layer, the 3D biological models, AR scene visualizations, bilingual labels, and audio narration assets.; and (4) the Evaluation Layer, comprising pre-test, post-test, and delayed post-test assessment instruments administered offline by the researcher.

The four-component applications AR Measurement Storybook, AR Life Science Storybook, AR Physical Science Storybook, and AR Number Concepts Storybook are architecturally independent. Each has its own Image Target database, its own 3D asset library, its own Narrative Manager logic, and its own evaluation instruments. No shared runtime dependency exists between components, ensuring that each can be deployed, tested, and evaluated without requiring any other component to be built or functional.

4.2 Component 2 Architecture - AR Life Science Storybook

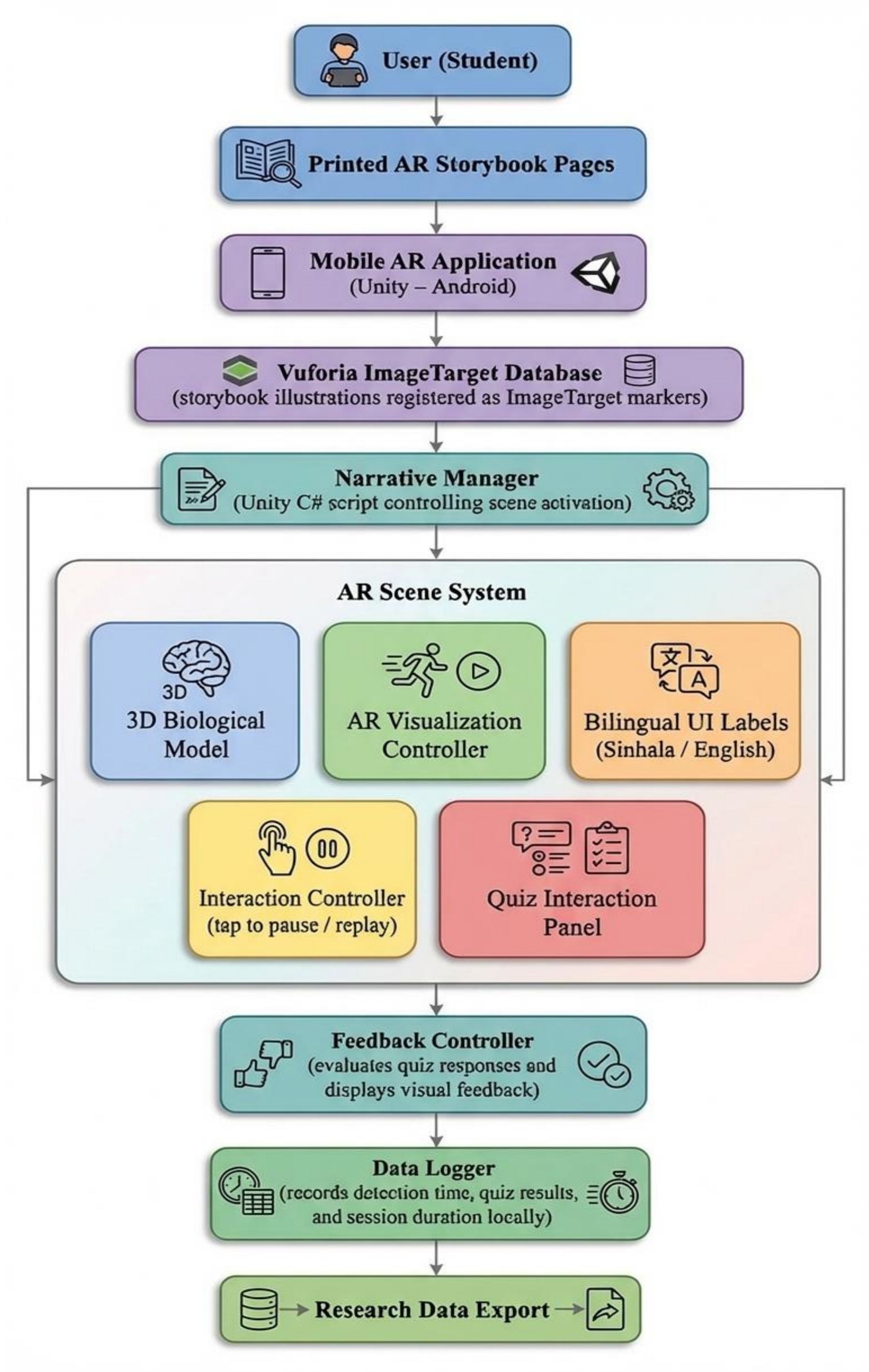

Figure 2 : Architecture - AR Life Science Storybook

The AR Life Science Storybook application is structured around the following architectural components:

Vuforia ImageTarget Database: Six-page illustrations from the printed storybook are registered as Vuforia ImageTarget markers. Each marker corresponds to one narrative stage of the story the seed, the germinating shoot, the food chain caterpillar stage, the bird-caterpillar predation stage, the decomposition stage, and the water cycle ecosystem stage.

Narrative Manager: A central C# script in Unity that listens for ImageTarget detection events from Vuforia and triggers the corresponding AR scene for the detected marker. The Narrative Manager enforces the page-specific narrative stage, ensuring that the AR content displayed always matches the story moment on the page the student is viewing.

AR Scene Components: Each of the six scenes contains (a) a 3D biological model created in Blender, (b) a Unity-based AR representing the biological process relevant to that page, (c) bilingual Sinhala and English UI labels rendered using Unity's TextMeshPro system, (d) a tap-to-pause and replay interaction controller, and (e) a quiz interaction panel that appears at the end of the animation sequence.

Feedback Controller: A C# component that evaluates student responses to the quiz interaction at the end of each scene, renders AR visual confirmation (green checkmark overlay for correct, gentle red

highlight for incorrect) and logs the response to a local device data file for later retrieval by the researcher.

Data Logger: A lightweight local file writer that records ImageTarget detection timestamps, quiz response accuracy, and session duration per student per scene. No data is transmitted over a network; all data is extracted by the researcher via device file export at the end of each session.

4.3 Data Flow

The data flow sequence is as follows: the student scans a storybook page with the device camera, Vuforia detects the ImageTarget and returns the marker ID and pose to Unity, the Narrative Manager maps the marker ID to the corresponding scene and triggers instantiation of the 3D AR scene, the student observes the AR visualization and interacts with the quiz at the scene conclusion, the Feedback Controller processes the response the Feedback Controller processes the response and renders visual feedback, and the Data Logger writes the session record to the local device storage. The student then turns to the next page and the cycle repeats.

4.4 Scalability and Security Considerations

Because the application operates entirely offline, there are no server-side security exposures. Student data is stored locally on the researcher's designated device, which is password-protected and not shared with students. The application architecture supports straightforward addition of new storybook pages by registering additional ImageTarget markers and creating corresponding Unity scenes, making the system extensible for future curriculum topics without requiring architectural changes. The application is designed to run on Android devices with a minimum of 2GB RAM, which represent the most common smartphone and tablet devices available in Sri Lankan schools., covering most affordable devices available in Sri Lanka.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The primary stakeholders of this system are Grade 4 and 5 primary school students in Sri Lanka aged 9 to 11, who are the direct users of the AR Life Science Storybook application. Secondary stakeholders include classroom teachers who facilitate the AR sessions, school administrators who approve and coordinate the research activities, parents and guardians who provide informed consent for student participation, and the research team responsible for evaluating outcomes.

5.2 Functional Requirements

1. **FR01 -** AR Storybook Scanning
   The application shall detect and track each of the six printed storybook page illustrations as Vuforia Image Target markers using the mobile device camera.
2. **FR02 -** AR Scene Rendering
   The application shall render the corresponding 3D AR biological process scene on a flat surface in front of the student upon detection of each Image Target marker.
3. **FR03 -** Narrative-Driven Sequencing
   The Narrative Manager shall ensure that the AR content displayed corresponds exclusively to the narrative stage of the storybook page being scanned.
4. **FR04 -** Animation Playback Controls
   The student shall be able to tap the screen to pause, resume, and replay each AR biological process visualization.
5. **FR05 -** Bilingual Labels
   All AR scene labels, instructions, and quiz questions shall be displayed in both Sinhala and English simultaneously.
6. **FR06 -** Quiz Interaction
   Each AR scene shall include a quiz interaction that prompts the student to identify what biological process has occurred, with selectable response options.
7. **FR07 -** Immediate Feedback
   The application shall provide immediate AR visual feedback: a green overlay for correct responses and a red highlight for incorrect responses following each quiz interaction.
8. **FR08 -** Data Logging
   The application shall log Image Target detection timestamps, quiz response accuracy per scene, and session duration to a local device file without requiring network access.
9. **FR09 -** Researcher Data Export
   The application shall store all session data in a format accessible to the researcher via the device file system without requiring internet connectivity.

5.3 Non-Functional Requirements

1. **Performance**
   The AR scene shall render within 3 seconds of Image Target detection on target Android devices.
2. **Usability**
   The application interface shall be operable without prior training by Grade 4–5 students aged 9 to 11, using touch interactions limited to tapping and slider dragging.
3. **Compatibility**
   The application shall function correctly on Android devices running version 8.0 and above without modification.
4. **Reliability**
   The application shall operate without internet connectivity throughout all student-facing sessions, with all required assets stored locally on the device.
5. **Accessibility**
   All text content shall be rendered at a minimum font size of 14pt within the AR overlay to ensure readability for young students at a typical holding distance of 30 to 50 centimeters.
6. **Security**
   No personally identifiable student data shall be transmitted over any network. All session data shall be stored locally on the researcher's device and protected by device-level access controls.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the AR Life Science Storybook is private primary schools and tuition centers in Sri Lanka that serve Grade 4 and 5 students preparing for the Grade 5 Scholarship Examination. The customer persona is a school principal or academic coordinator who is seeking to differentiate the school's science instruction with technology-enhanced learning resources and who has access to a small fleet of shared Android tablets. Secondary markets include Sri Lankan educational publishers seeking to enhance printed textbook products with AR content, and international education technology companies seeking validated AR learning frameworks for South Asian markets.

6.2 Value Proposition

The AR Life Science Storybook provides an evidence-based, curriculum-aligned, bilingual AR learning experience that demonstrably improves student understanding of life science topics compared to existing textbook instruction. It is the only AR science learning tool designed specifically for the Sri Lankan primary curriculum, the only bilingual Sinhala and English AR science educational product, and the only AR system to use locally familiar biodiversity and culturally grounded narrative settings to ground biological concepts in students lived experience. The storybook's illustrated artistic narrative also makes this the only AR science product in Sri Lanka to formally embed Art within the STEAM learning experience, using visual storytelling as a pedagogical bridge to scientific understanding.

6.3 Revenue Model

The system is designed to support a B2B institutional licensing model, in which schools or tuition centers purchase a per-device annual license that includes the application, the printed storybook, and access to curriculum-aligned assessment tools. A freemium consumer model in which a limited version of the application is available on the Google Play Store with in-app purchase to unlock the full storybook set provides an additional revenue stream targeting parents of Grade 4 and 5 students.

6.4 Pricing Strategy

Based on comparable AR educational applications in the South Asian market, an institutional license price of LKR 2,500 to LKR 3,500 per device per academic year is estimated to be commercially viable while remaining affordable for mid-range private schools. Consumer pricing for the full in-app purchase is estimated at LKR 990.

6.5 Competitive Advantage

No competing product offers AR life science content specifically aligned with the Sri Lankan Grade 4 and 5 Parisaraya curriculum in Sinhala and English. All globally available AR science apps address generic or Western curricula, use English only, and do not feature local biodiversity. This combination of curriculum specificity, bilingual accessibility, and cultural grounding creates a defensible competitive moat in the Sri Lankan educational technology market.

6.6 Intellectual Property Considerations

The narrative storybook text, 3D biological models, and Unity application code developed for this project are original works eligible for copyright protection. The research team will retain intellectual property rights for all assets created during the project. The Vuforia SDK is used under Vuforia's standard licensing terms, which permit commercial deployment. Unity is used under Unity's standard commercial license. A provisional patent application may be considered for the Narrative Manager AR sequencing methodology if the evaluation demonstrates a novel and non-obvious technical contribution.

7. BUDGET AND JUSTIFICATION

Category	Estimated Cost (LKR)	Justification
Hardware	35,000	Required for AR development and testing on target device specifications.
Software	0	Free Personal tier sufficient for academic non-commercial use during development phase.
Software	0	Vuforia basic license is available at no cost for academic and non-commercial development.
Software	0	Open-source, no licensing cost.
Printing	8,000	Estimated cost for printing 10 research copies of the 6-page bilingual storybook for evaluation sessions.
Cloud	0	Free tier sufficient for research data volumes.
Human Effort	280,000	Estimated value of researcher development and evaluation effort. *Confirm exact figure with supervisor.
Miscellaneous	LKR 5,000	Travel and logistics for school-based data collection sessions.
TOTAL	**LKR 328,000**	

Table 1: Budget and Justification

The estimated monetary value for human effort is based on an hourly rate of LKR 500 for undergraduate research development work in Sri Lanka. This figure should be confirmed with the supervisor before finalizing the budget.

8. WORK BREAKDOWN STRUCTURE (WBS)

1. **Project Planning and Requirements Definition**
   Define system requirements, select development tools, and establish the overall project.
2. **Literature Review and Research Gap Analysis**
   Review existing AR learning research and identify gaps related to primary science education.
3. **Storybook Narrative Design and Layout**
   Design the bilingual AR storybook narrative and page layouts for AR marker integration.
4. **ImageTarget Database Setup**
   Register storybook illustrations as Vuforia ImageTarget markers and validate detection quality.
5. **3D Biological Model Creation**
   Develop optimized 3D models representing life science processes using Blender.
6. **Unity AR Scene Development**
   Create interactive AR scenes that visualize biological processes, with labels, narration, and learning interactions.
7. **Narrative Manager and System Logic Development**
   Implement scripts to manage AR scene activation, story progression, and system control.
8. **Bilingual User Interface Integration**
   Add Sinhala and English labels, instructions, and UI components for accessibility.
9. **System Testing and Optimization**
   Test AR functionality, marker detection, interaction features, and overall system performance.
10. **Storybook Printing and Research Preparation**
    Print finalized storybook pages and prepare materials for classroom research sessions.
11. **Evaluation and Data Collection**
    Conduct pre-tests, AR learning sessions, and post-tests to measure learning outcomes.
12. **Data Analysis and Dissertation Completion**
    Analyze collected data and compile the final research dissertation for submission.

9. GRANTT CHART

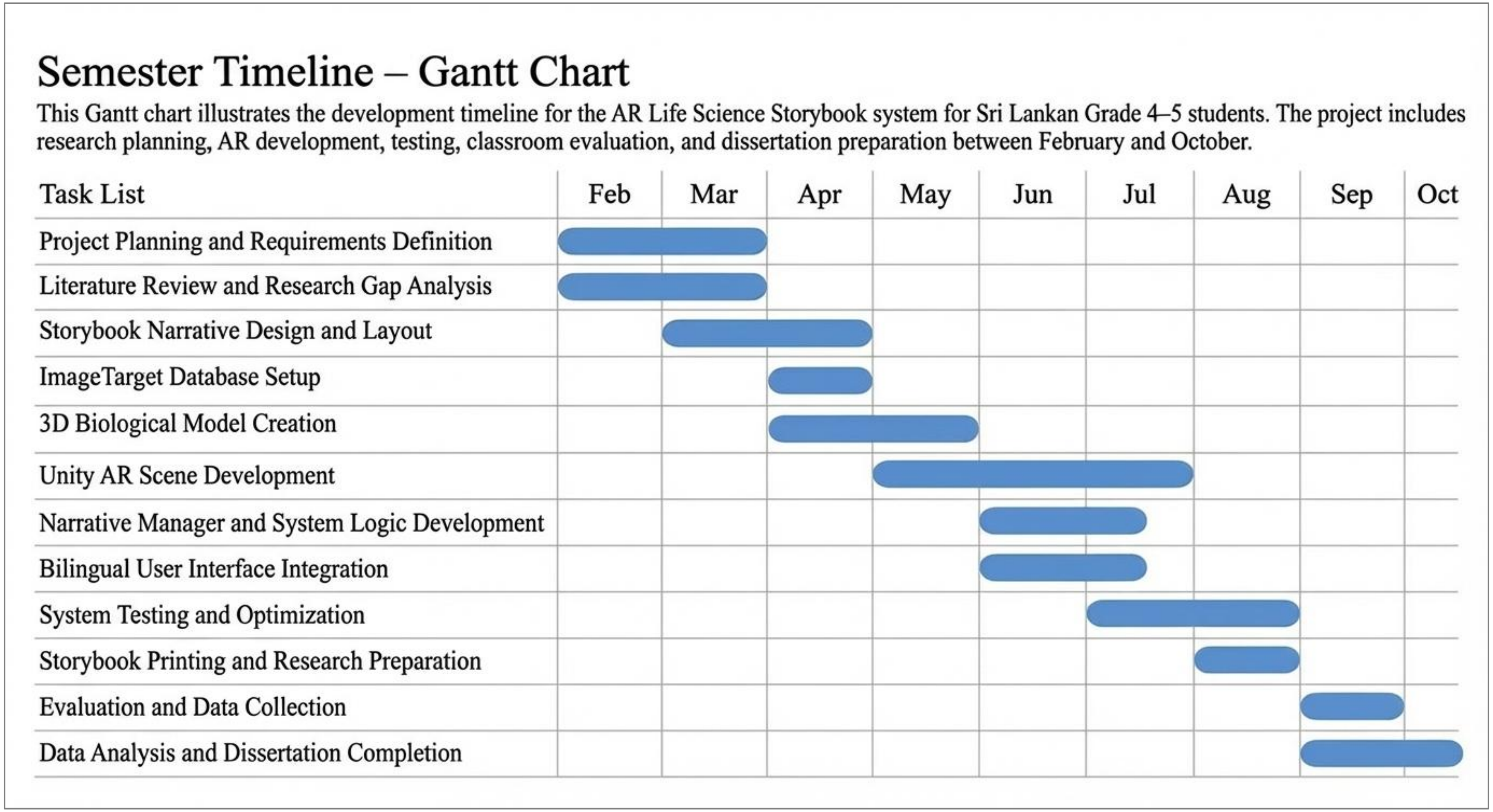

Figure 3 : Semester Timeline - Gant Chart

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